

Lecture 05: Matching

EEFE 530

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Schedule

Roadmap

Today's lecture:

- Review identification under selection on observables
- Introduce matching as a way to construct counterfactuals
- Discuss common matching estimators
- Highlight the role of overlap and common support
- Compare matching to regression adjustment

Learning Objectives

By the end of this lecture, you should be able to:

- Explain when selection on observables is a plausible assumption
- Describe how matching constructs counterfactual outcomes
- Distinguish between different matching estimators
- Explain why common support is required for identification
- Compare matching and regression as adjustment strategies

Upcoming

- Problem Set 3 on matching
- Instrumental variables
- Difference-in-difference

Reading

Instrumental Variables

- Mixtape Ch. 7
- Effect Ch. 19
- *MHE* Ch. 4

Regression

Regression

Recall: Regression serves as the conditional expectation function (CEF) where:

$$y_i = \mathbf{X}_i' \boldsymbol{\beta} + \epsilon_i$$

$$E[Y_i | X_i] = \mathbf{X}_i' \boldsymbol{\beta}$$

$$\boldsymbol{\beta} = E[\mathbf{X}_i \mathbf{X}_i']^{-1} E[\mathbf{X}_i Y_i]$$

In **simple linear regression** (an intercept and one regressor x_i),

$$\beta_1 = \frac{\text{Cov}(Y_i, x_i)}{\text{Var}(x_i)} \quad \beta_0 = E[Y_i] - \beta_1 E[x_i]$$

For **multivariate regression**, the coefficient on the k^{th} regressor x_{ki} is

$$\beta_k = \frac{\text{Cov}(Y_i, \tilde{x}_{ki})}{\text{Var}(\tilde{x}_{ki})}$$

where $\tilde{\mathbf{x}}_{ki}$ is the residual from a regression of $\mathbf{x}_{ki}$ on all other covariates.

Regression

Great! Now we have a conditional expectation of Y_i .

...but we care about the interpretation of β . When/how can we interpret β as the *causal* effect of X on Y ?

For traditional regression we need the **conditional independence assumption (CIA)**, also called *selection on observables*.

Q: What does CIA mean in words?

A: Conditional on some set of covariates X_i , selection bias disappears.

More formally Conditional on X_i , potential outcomes (Y_{0i}, Y_{1i}) are independent of treatment status (D_i) .

$$\{Y_{0i}, Y_{1i}\} \perp\!\!\!\perp D_i | X_i$$

Regression & CIA

Let's formally link regression to the CIA.

We have two linear, population models for the returns to schooling ($\mathbf{Y}_i$ is earnings and $\mathbf{s}_i$ is years of schooling.)

$$\mathbf{Y}_i = \alpha + \rho \mathbf{s}_i + \eta_i \quad (1)$$

$$\mathbf{Y}_i = \alpha + \rho \mathbf{s}_i + \mathbf{X}_i' \boldsymbol{\gamma} + \nu_i \quad (2)$$

$\hat{\rho}$ in Model (1) does not have a causal interpretation if there is selection bias.

Q: What is the concern of selection bias for model (1) in words?

A: People with more schooling would have earned more even if they hadn't gone to school (better ability, more motivation, etc.)

For model (2), we can interpret $\hat{\rho}$ causally if $\mathbf{Y}_i \perp\!\!\!\perp \mathbf{s}_i \mid \mathbf{X}_i$ (CIA).

In other words, the CIA says that our **observable vector** $\mathbf{X}_i$ **must explain all of correlation between** $\mathbf{s}_i$ **and** η_i .

Omitted variable bias

We can define the bias from *not* controlling for relevant (i.e. correlated with **Y** and **X**) variables using the omitted-variable bias (OVB) formula.

Let's assume we have data on all the factors that are correlated with both education and earnings. We'll call this: (**A_i**) (family background, motivation, intelligence, etc.).

$$Y_i = \alpha + \beta s_i + v_i \quad (1)$$

$$Y_i = \pi + \rho s_i + A_i' \gamma + e_i \quad (2)$$

What happens if we can't get data on **A_i** and opt for (1)?

$$\frac{\text{Cov}(Y_i, s_i)}{\text{Var}(s_i)} = \rho + \gamma' \delta_{As}$$

where δ_{As} are coefficients from regressing **A_i** on **s_i** .

Omitted variable bias

Interpretation

Our two regressions

$$Y_i = \alpha + \beta s_i + v_i \quad (1)$$

$$Y_i = \pi + \rho s_i + A_i' \gamma + e_i \quad (2)$$

will yield the same estimates for the returns to schooling

$$\frac{\text{Cov}(Y_i, s_i)}{\text{Var}(s_i)} = \rho + \gamma' \delta_{As}$$

if **(a)** schooling is uncorrelated with ability ($\delta_{As} = 0$) or

(b) ability is uncorrelated with earnings, conditional on schooling ($\gamma = 0$).

Additionally, we can think about the direction of the bias from model (1) if we have can't get the data on $\mathbf{A}_i$, but have some idea about the signs of δ_{As} and γ .

Matching

Goals

Let's forget our little diversion about the regression, and come back to the fundamental problem of causal inference.

1. We don't observe both $\mathbf{Y}_{1i}$ and $\mathbf{Y}_{0i}$ simultaneously.
2. We may be worried about selection bias when comparing $E[\mathbf{Y}_{1i} | \mathbf{D}_i = 1]$ with $E[\mathbf{Y}_{0i} | \mathbf{D}_i = 0]$. In particular we are worried $E[\mathbf{Y}_{0i} | \mathbf{D}_i = 1] \neq E[\mathbf{Y}_{0i} | \mathbf{D}_i = 0]$.

Most empirical strategies aim to estimate $\mathbf{Y}_{0i}$ for treated individuals—the unobservable counterfactual for the treatment group.

Matching estimators are one way to generate these counterfactuals by matching treated observations with “similar” control observations. Formally we still need to be able to make a CIA assumption.

Exact Matching

If we believe the CIA $(Y_{1i}, Y_{0i}) \perp\!\!\!\perp D_i | X_i$, then we have a good counterfactual: $E[Y_{0i} | X_i, D_i = 0] = E[Y_{0i} | X_i, D_i = 1]$

- We don't observe $E[Y_{0i} | X_i, D_i = 1]$
- We do observe $E[Y_{0i} | X_i, D_i = 0]$

We can estimate the ATT (τ_{ATT}) by simply taking the difference of treated and control observations with the same X values:

$$\begin{aligned}\tau_{ATT} &= E\{E[Y_{1i} | X_i, D_i = 1] - E[Y_{0i} | X_i, D_i = 0] | D_i = 1\} \\ &= E[\tau_X | D_i = 1]\end{aligned}$$

where $\tau_x \equiv E[Y_i | X_i, D_i = 1] - E[Y_i | X_i, D_i = 0]$

Exact Matching

Exact matching has a relatively simple and elegant aggregation that can be thought of as weighting each of the mean differences within cells of covariate values.

Recall that $\tau_{ATT} = E[\tau_X | D_i = 1]$.

We can denote the mean difference at a particular value of $X_i = x$ as τ_x , which allows us to define:

$$\tau_{ATT} = E[(Y_{1i} - Y_{0i}) | D_i = 1] = \sum_x \tau_x P(X_i = x | D_i = 1)$$

$P(X_i = x | D_i = 1)$ is the probability mass function for X_i given that $D_i = 1$.

Exact Matching

We can also generalize the treatment effect to the population by removing the conditioning on $D_i = 1$.

$$\begin{aligned}\tau_{ATE} &= E\{E[Y_{1i}|X_i, D_i = 1] - E[Y_{0i}|X_i, D_i = 0]\} \\ &= E[Y_{1i} - Y_{0i}] = \sum_x \tau_x P(X_i = x)\end{aligned}$$

In words: now we are weighting by the full marginal distribution of X_i instead of the distribution of the treated population.

Exact Matching

Now is a good time to take a step back and think about the feasibility/practicality of exact matching.

1. We need to have a discrete valued covariates.
2. We need overlap among covariates (we need treated and untreated observations to have the same values of $\mathbf{X}_i$)
3. We need the CIA to hold given our available data.

Exact matching provides a nice simple example of how matching is operationalized, but these properties may not hold in practice.

Next, we'll generalize matching to a method of finding counterfactuals based on observables and weighting the difference between treated and counterfactual control observations.

Quick note: Coarsened exact matching makes exact matching feasible with continuous variables by creating bins (e.g. quantiles) of the continuous variables and matching within those bins.

Context: Regression vs. Matching

MHE shows how certain regression models can be thought of as matching estimators.

In particular, saturated in $\mathbf{X}_i$ models (models that include dummies for every value of $\mathbf{X}_i$) are equivalent to exact matching with different weights.

With some math (see *MHE* p74-75) the regression coefficient on a dummy treatment variable with saturated $\mathbf{X}_i$ can be written as:

$$E[Y_{1i} - Y_{0i} | D_i = 1] = \tau_{ATT,R} = \frac{E[\sigma_D^2(\mathbf{X}_i)\tau_x]}{E[\sigma_D^2(\mathbf{X}_i)]}$$

where $\sigma_D^2(\mathbf{X}_i)$ is the conditional variance of D_i given $\mathbf{X}_i$.

Given that D_i is a dummy variable we have

$$\tau_{ATT,R} = \frac{\sum_x \tau_x [P(D_i=1|X_i=x)(1-P(D_i=1|X_i=x))]P(X_i=x)}{\sum_x [P(D_i=1|X_i=x)(1-P(D_i=1|X_i=x))]P(X_i=x)}$$

Context: Regression vs. Matching

So we have:

$$\tau_{ATT,R} = \frac{\sum_x \tau_x [P(D_i=1|X_i=x)(1-P(D_i=1|X_i=x))]P(X_i=x)}{\sum_x [P(D_i=1|X_i=x)(1-P(D_i=1|X_i=x))]P(X_i=x)}$$

$$\tau_{ATT,M} = \sum_x \tau_x P(X_i = x | D_i = 1)$$

which also equals

$$\tau_{ATT,M} = \frac{\sum_x \tau_x P(D_i=1|X_i=x)P(X_i=x)}{\sum_x P(D_i=1|X_i=x)P(X_i=x)}$$

given that $P(X_i = x | D_i = 1) = \frac{P(D_i=1|X_i=x)P(X_i=x)}{P(D_i=1)}$

Context: Regression vs. Matching

So the ATT estimated through matching and regression are both taking weighted averages of the difference in means at different covariate value cells.

For $\tau_{ATT,M}$ we weight the difference in means by the conditional distribution of $\mathbf{X}_i | \mathbf{D}_i = \mathbf{1}$ - the characteristics of the treated population.

For $\tau_{ATT,R}$ we weight by the conditional variance of $\mathbf{D}_i$.

Some key insights:

General setup

We want to construct a counterfactual for each individual with $\mathbf{D}_i = \mathbf{1}$.

The counterfactual for i should only use individuals that match $\mathbf{X}_i$.

Let there be N_T treated individuals and N_C control individuals. We want:

- N_T sets of weights
- with N_C weights in each set

$$w_i(j) \quad (i = 1, \dots, N_T; j = 1, \dots, N_C)$$

More formally

If our estimated counterfactual for treated individual i is

$$\widehat{Y}_{0i} = \sum_j w_i(j) Y_j$$

then our estimated treatment effect (for individual i) is

$$\hat{\tau}_i = Y_{1i} - \widehat{Y}_{0i} = Y_{1i} - \sum_j w_i(j) Y_j$$

a generic matching estimator for the treatment effect on the treated is

$$\hat{\tau}_M = \frac{1}{N_T} \sum_{i \in (D=1)} \left(Y_{1i} - \widehat{Y}_{0i} \right) = \frac{1}{N_T} \sum_{i \in (D=1)} \left(Y_{1i} - \sum_{j \in (D=0)} w_i(j) Y_j \right)$$

Weighting

So all we need is those weights and we're done... great.

Q: Where does one find the weights?

A: You've got options, but you need to choose carefully/responsibly.

E.g., if $w_i(j) = \frac{1}{N_C}$ for all (i, j) , then we're back to a difference in means.
We can't use the conditional independence assumption because we're not conditioning on any observables.

Gameplan: Choose weights $w_i(j)$ that indicate *how close* $\mathbf{X}_j$ is to $\mathbf{X}_i$.

Proximity

Our weights $w_i(j)$ should be a measure of *how close* $\mathbf{X}_j$ is to $\mathbf{X}_i$.

If $\mathbf{X}$ is *discrete*, then we can consider equality, *i.e.*, $w_i(j) = 1\{\mathbf{X}_i = \mathbf{X}_j\}$, scaling as necessary to get $\sum_j w_i(j) = 1$.

This is essentially what we did with exact matching. Assign a uniform constant weight to all observations that had the exact same covariate values of a treatment observation and assign a weight of zero to all other observations.

Proximity

Our weights $w_i(j)$ should be a measure of **how close** $\mathbf{X}_j$ is to $\mathbf{X}_i$.

If $\mathbf{X}$ is *continuous*, then we need *proximity* rather than *equality*.

Nearest-neighbor matching chooses the single closest control observation using the Euclidean distance between $\mathbf{X}_i$ and $\mathbf{X}_j$, i.e.,

$$d_{i,j} = (\mathbf{X}_i - \mathbf{X}_j)' (\mathbf{X}_i - \mathbf{X}_j)$$

Proximity

Our weights $w_i(j)$ should be a measure of **how close** $\mathbf{X}_j$ is to $\mathbf{X}_i$.

If $\mathbf{X}$ is *continuous*, then we need *proximity* rather than *equality*.

Nearest-neighbor matching with Mahalanobis distance chooses the single closest control using *Mahalanobis* distance between $\mathbf{X}_i$ and $\mathbf{X}_j$, i.e.,

$$d_{i,j} = (\mathbf{X}_i - \mathbf{X}_j)' \Sigma_X^{-1} (\mathbf{X}_i - \mathbf{X}_j)$$

where Σ_X^{-1} is the covariance matrix of $\mathbf{X}$.

- **Estimator:** $\hat{\tau}_M = \frac{1}{N_T} \sum_i \hat{\tau}_i$ where $(\hat{\tau}_i = Y_{1i} - Y_{0j}^i)$
- Produces causal estimates if CIA is valid *and* we have sufficient overlap.
- Does not suffer from arbitrary choices of units.

More neighbors?

Why limit ourselves to a **single** “best” match?

If we’re going to let a function/algorithm choose the *nearest* match, can’t we also let the function/algorithm choose *how many* matches?

Furthermore, if $N_C \gg N_T$, we’re throwing away *a lot* of potentially useful information.

We could instead use this information and be more efficient.

More neighbors!

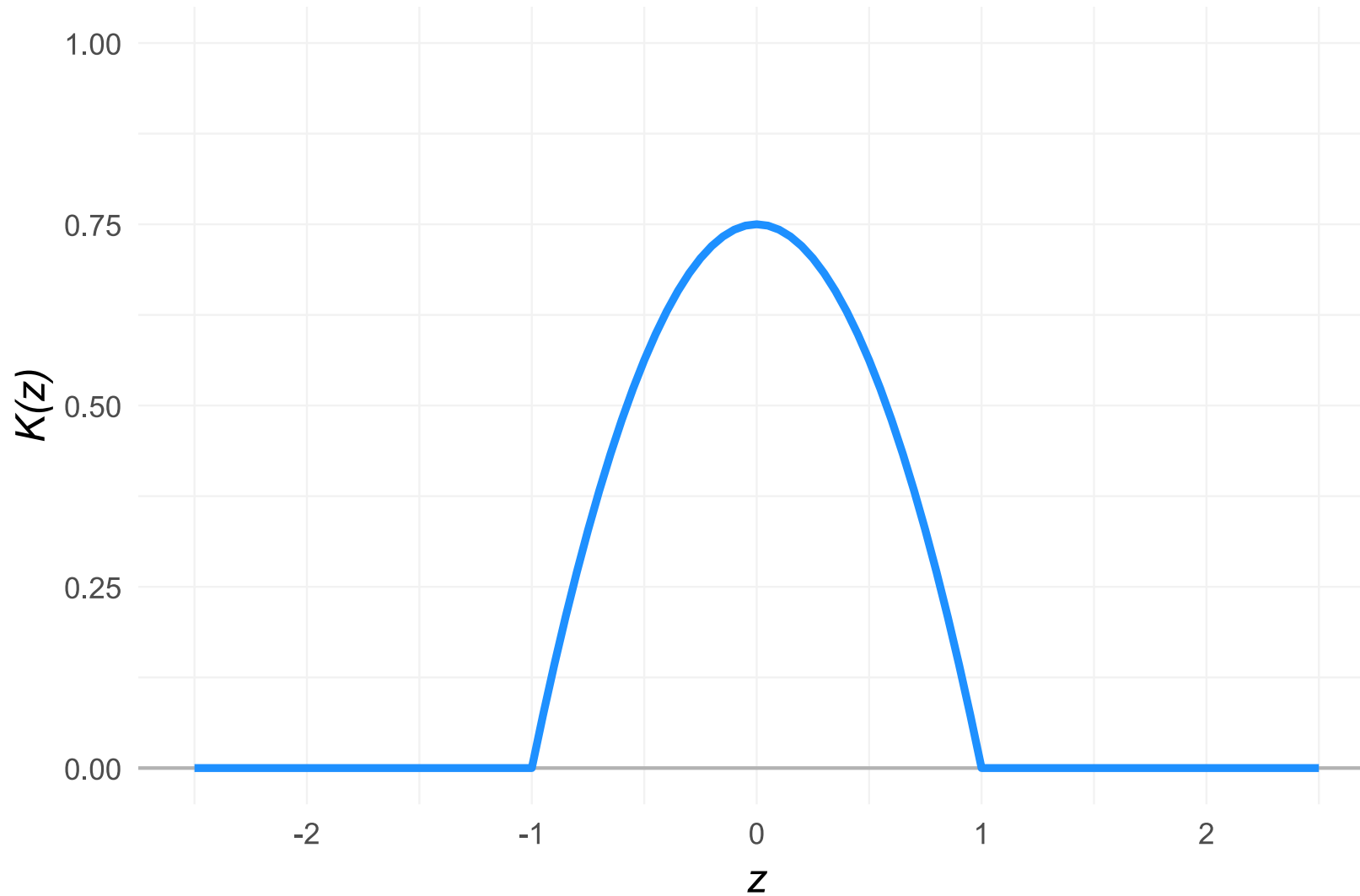
Kernel matching gives positive weight to all control observations within some **bandwidth** h , with higher weight for closer matches determined by some **kernel function** $K(\cdot)$,

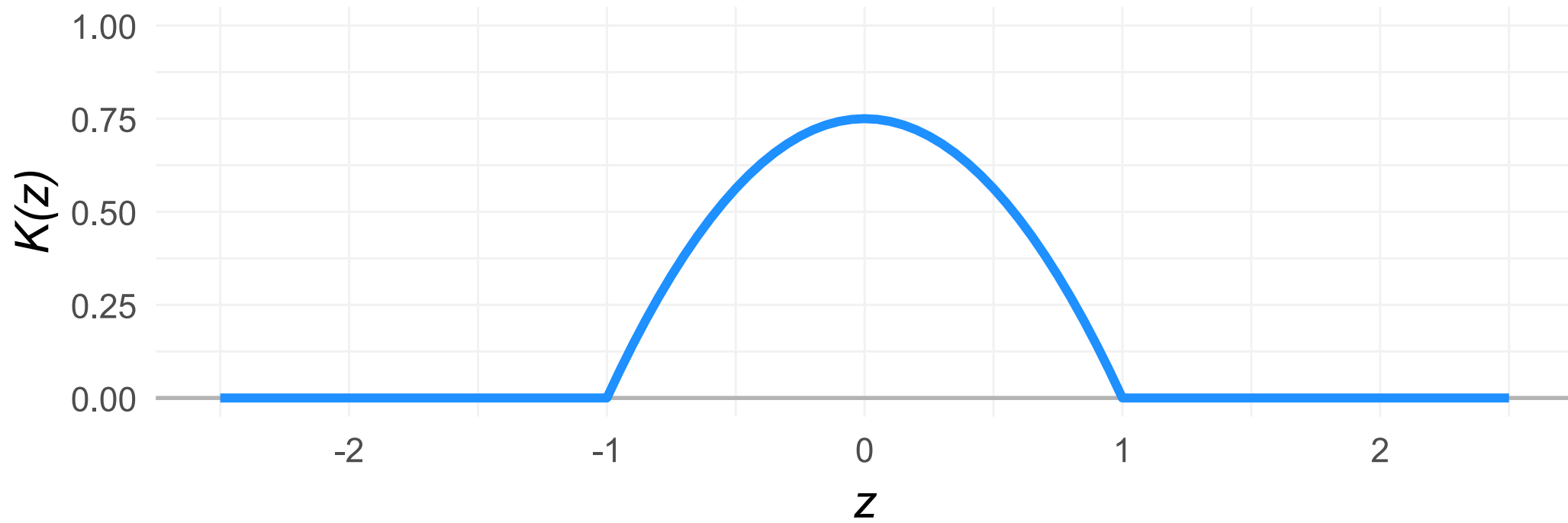
$$w_i(j) = \frac{K\left(\frac{X_j - X_i}{h}\right)}{\sum_{j \in (D=0)} K\left(\frac{X_j - X_i}{h}\right)}$$

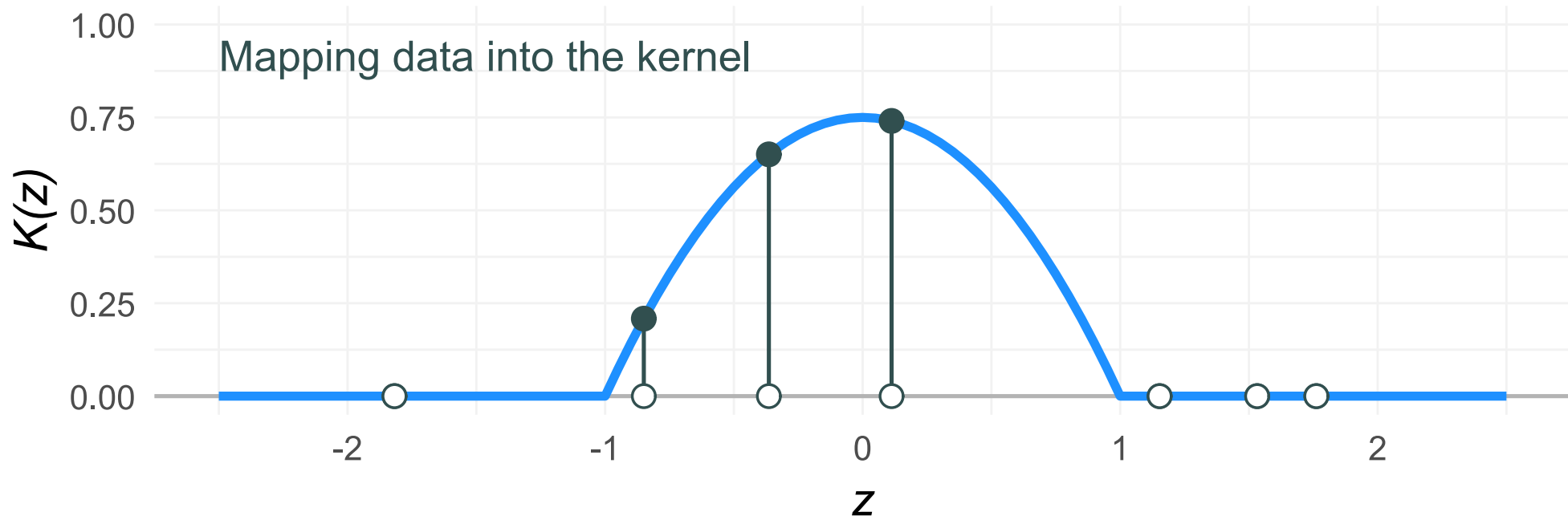
Example: The *Epanechnikov kernel* is defined as

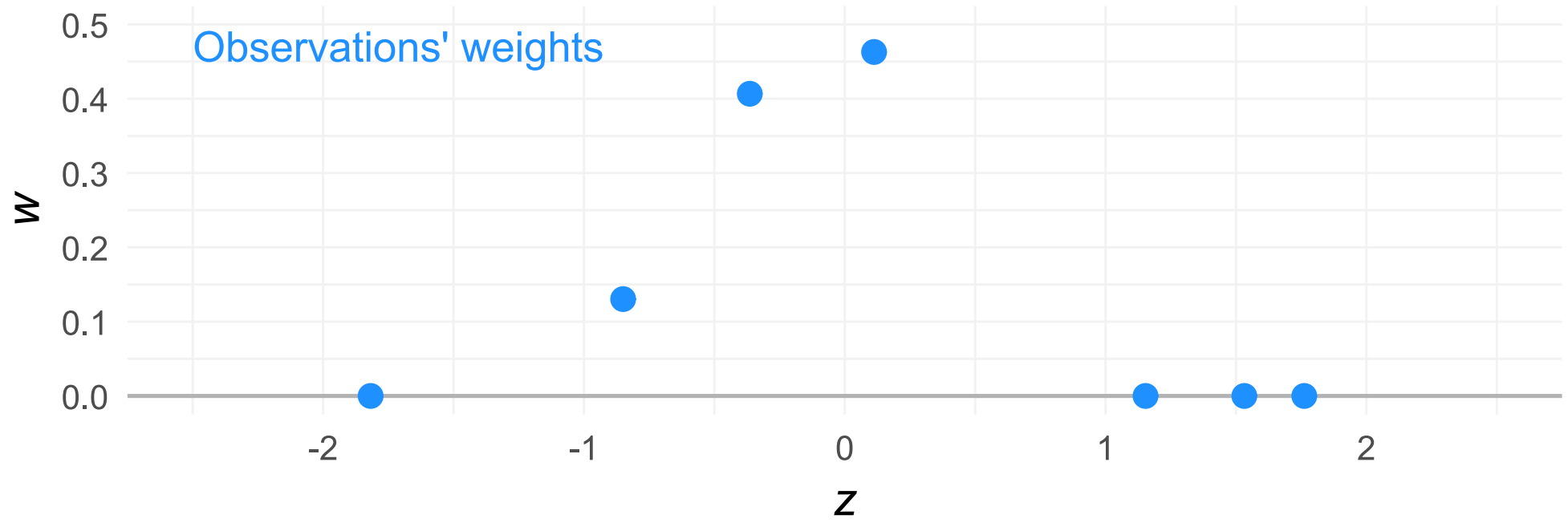
$$K(z) = \frac{3}{4} (1 - z^2) \times 1\{|z| < 1\}$$

The Epanechnikov kernel $K(z) = \frac{3}{4} (1 - z^2) \times \mathbf{1}\{|z| < 1\}$

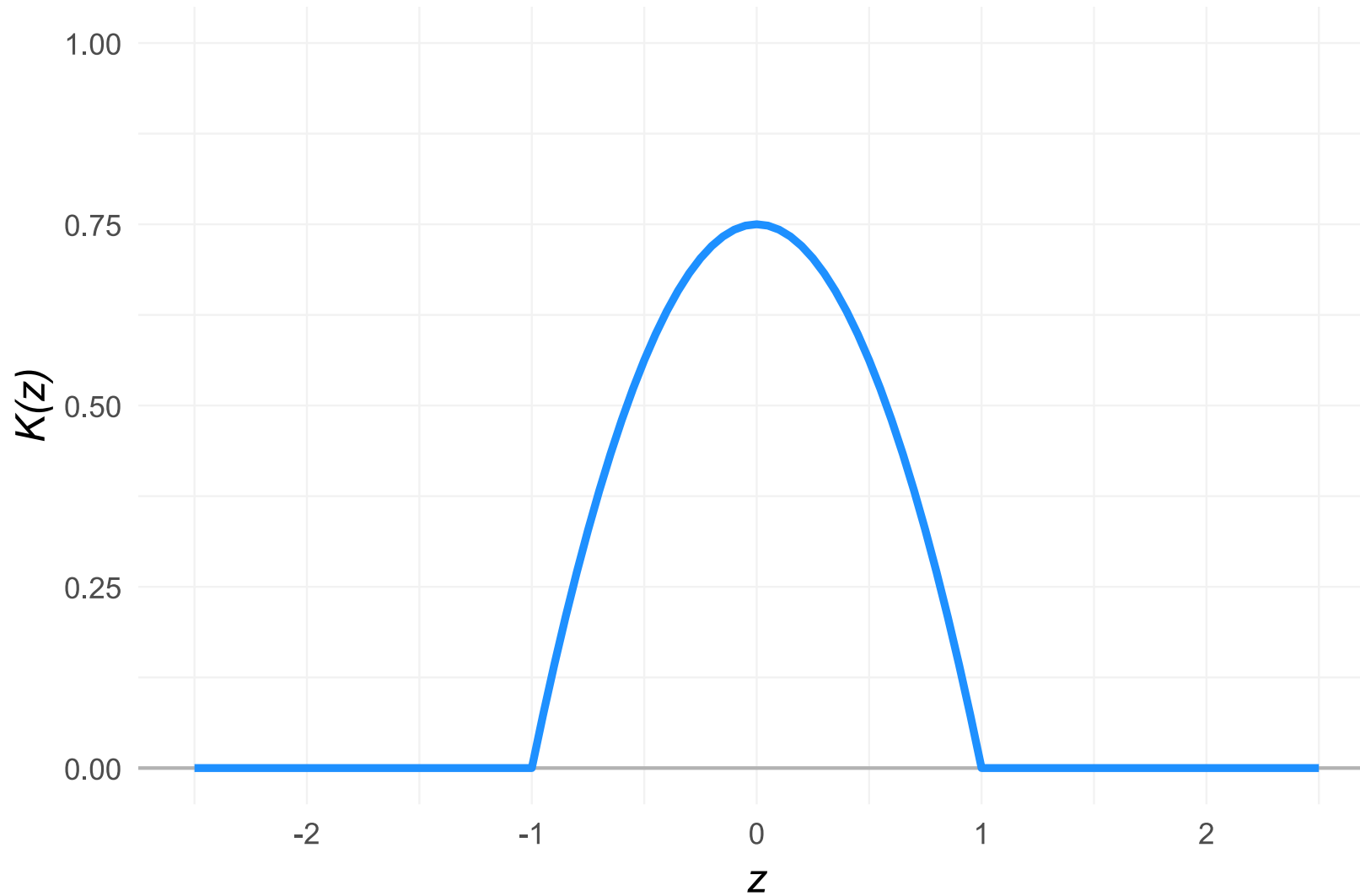




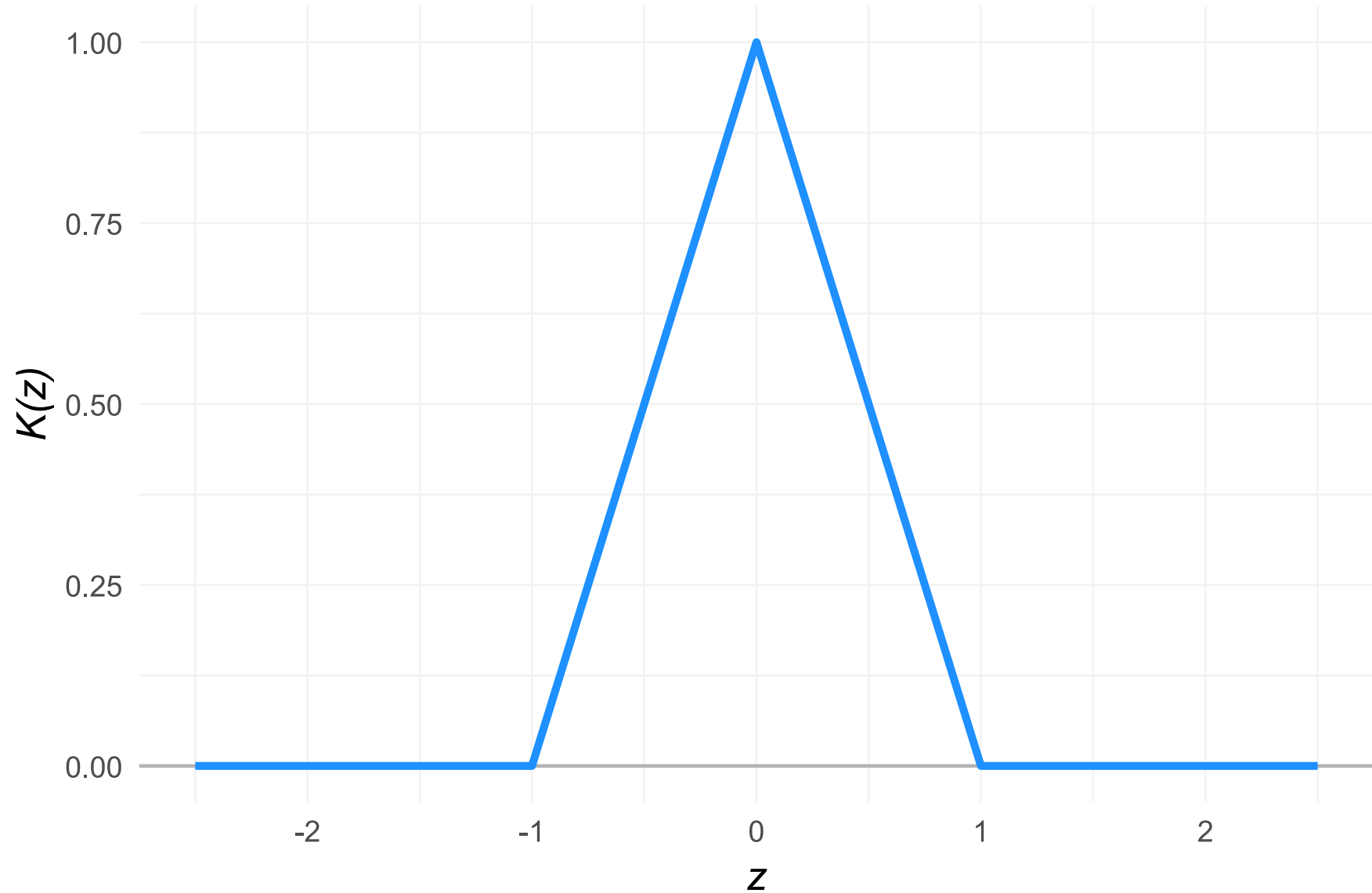




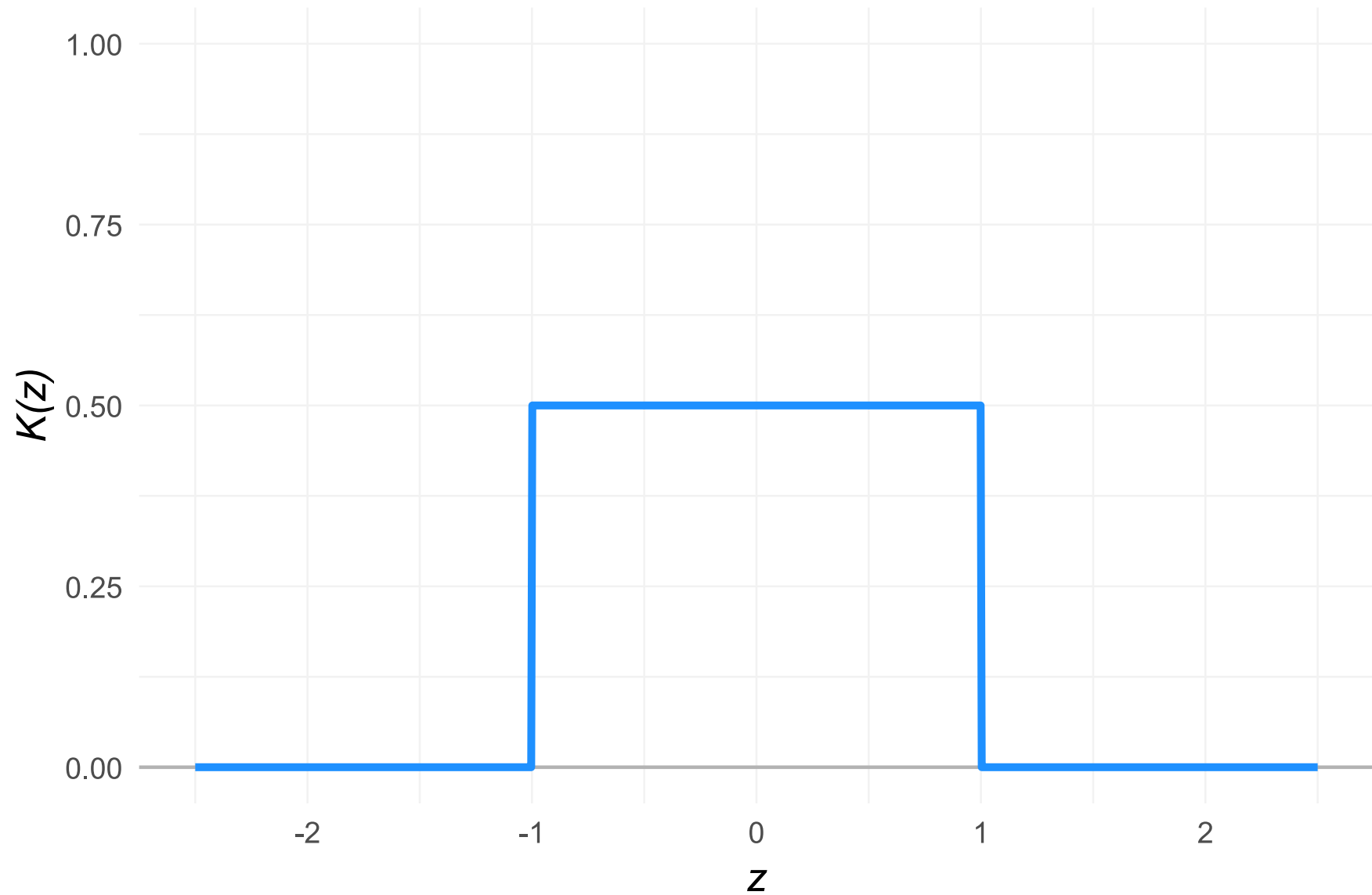
The Epanechnikov kernel $K(z) = \frac{3}{4} (1 - z^2) \times \mathbf{1}\{|z| < 1\}$



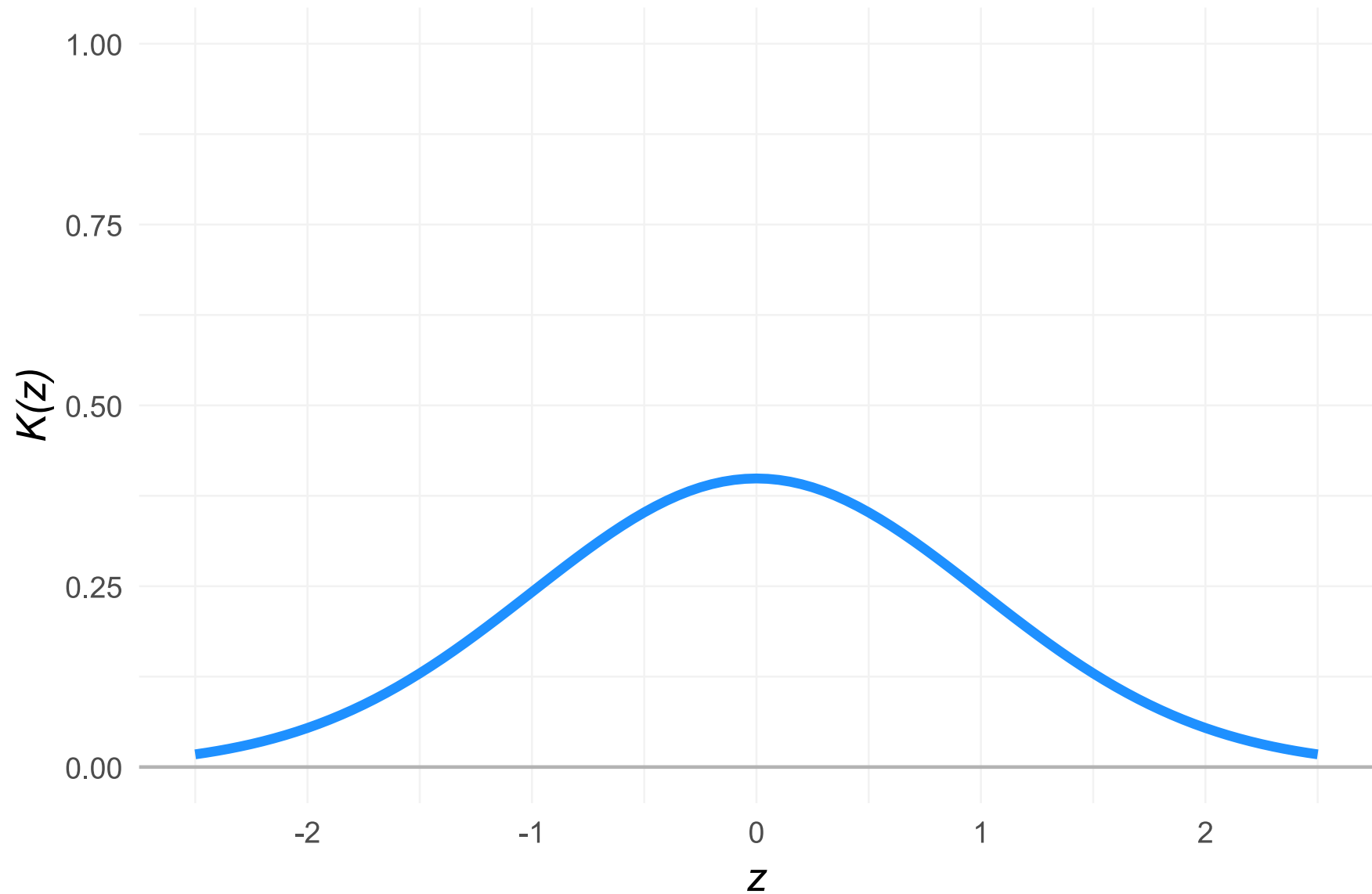
The Triangle kernel $K(z) = (1 - |z|) \times 1\{|z| < 1\}$



The Uniform kernel $K(z) = \frac{1}{2} \times 1\{|z| < 1\}$



The Gaussian kernel $K(z) = (2\pi)^{-1/2} \exp(-z^2/2)$



Kernels

Aside

Kernel functions are good for more than just matching.

You will most commonly see/use them smoothing out densities—providing a smooth, moving-window average.

E.g., R's (**ggplot2**'s) smooth, density-plotting function **geom_density()**.

geom_density() defaults to **kernel = "gaussian"**, but you can specify many other kernel functions (including **"epanechnikov"**).

You can also change the **bandwidth** argument. The default is a bandwidth-choosing function called **bw.nrd0()**.

Matching

Adding neighbors

As we add more neighbors—either moving from **1** to $n > 1$ or increasing our bandwidth—we potentially increase the efficiency of our estimator.

We need to **be careful not to add *too many* controls** for each treated i .

CIA requires that we're actually conditioning on the observables—it does not allow us to take a simple average across all control observations.

This has the flavor of a bias-variance tradeoff.

The curse of dimensionality

It turns out kernel- and bandwidth-selection are not our biggest enemies.

As the dimension of **X** expands (matching on more variables), it becomes **harder and harder to find a nice, close control** for each treated unit.

We need a way to shrink the dimensionality of **X**.

Propensity-score methods

Setup

Let's begin with two assumptions—one old and one new.

1. **Conditional independence:** $(Y_{0i}, Y_{1i}) \perp\!\!\!\perp D_i | X_i$
2. **Overlap:** $0 < \Pr(D_i = 1 | X_i) < 1$

We can estimate an average treatment effect by conditioning on X_i .

However, overlap may fail if the dimensions of X are large and N is finite.

Propensity scores provide a solution to this challenge

The trick

It turns out that if $(Y_{0i}, Y_{1i}) \perp\!\!\!\perp D_i | X_i$, then we actually only need to match/condition on $p(X_i) = E[D_i | X_i]$.

$p(X_i)$ is the **propensity score**

the probability of treatment given X_i .

Propensity-score theorem If $(Y_{0i}, Y_{1i}) \perp\!\!\!\perp D_i | X_i$, then $(Y_{0i}, Y_{1i}) \perp\!\!\!\perp D_i | p(X_i)$.

This theorem extends our CIA to a one-dimensional score, avoiding the curse of dimensionality.

Intuition

Q: What's going on here?

$\mathbf{X}_i$ carries way more information than $p(\mathbf{X}_i)$, so how can we still get conditional independence of treatment by only conditioning on $p(\mathbf{X}_i)$?

A₁ : Conditional independence of treatment isn't about extracting all of the information possible from $\mathbf{X}_i$. We actually only using $\mathbf{X}_i$ such that $\mathbf{D}_i | \mathbf{X}_i$ is independent of (Y_{0i}, Y_{1i}) .

A₂ : Back to our main concern: **selection bias**. People select into treatment. If $\mathbf{X}_i$ says two people were equally likely to be treated, and if $\mathbf{X}_i$ explains all of selection (CIA), then there cannot be selection bias between these two people.

Another way of saying this is that the information we care about in $\mathbf{X}_i$ pertains to the probability (propensity) of being treated, which is $p(D_i = 1 | \mathbf{X}_i) = p(\mathbf{X}_i)$.

Estimation

So where do propensity scores come from?

We estimate them—and there are a lot of ways to do that.

1. Flexible (*i.e.*, interactions) logit specification
2. Kernel regression (remember kernel functions?)
3. Many others—machine learning, series-logit estimator, *etc.*

Q: Can we just use plain OLS (linear probability model)?

A: Sort of. Think about FWL. This route is going to be the same as a regression conditioning on $\mathbf{X}_i$.

Estimation

From *MHE* (p. 83)

Question

A big question here is how to best model and estimate $p(\mathbf{X}_i)$...

Answer

The answer to this is inherently application-specific. A growing empirical literature suggests that a logit model for the propensity score with a few polynomial terms in continuous covariates works well in practice...

See Caliendo & Kopeinig (2008), *SOME PRACTICAL GUIDANCE FOR THE IMPLEMENTATION OF PROPENSITY SCORE MATCHING*

Application

So you have some estimated propensity scores $\hat{p}(\mathbf{X}_i)$. What next?

Option 1: Conditioning via regression

$$\mathbf{Y}_i = \alpha + \delta \mathbf{D}_i + \beta \mathbf{X}_i + u_i \quad (1)$$

Option 1a: Use a **regression to condition** on $p(\mathbf{X}_i)$, i.e.,

$$\mathbf{Y}_i = \alpha + \delta \mathbf{D}_i + \beta p(\mathbf{X}_i) + u_i \quad (1a)$$

Option 1b: If we think treatment effects are heterogeneous and may covary with $\mathbf{X}$, then we might want to also **interact** treatment with $p(\mathbf{X}_i)$, i.e.,

$$\mathbf{Y}_i = \alpha + \delta_1 \mathbf{D}_i + \delta_2 \mathbf{D}_i p(\mathbf{X}_i) + \beta p(\mathbf{X}_i) + u_i \quad (1b)$$

Heterogeneity with regression

Let's think a bit more about heterogeneous treatment effects in this setting.

$$\begin{aligned}Y_{0i} &= \alpha + \beta X_i + u_i \\Y_{1i} &= Y_{0i} + \delta_1 + \delta_2 X_i\end{aligned}$$

i.e., the treatment effect depends upon X_i .

$$\begin{aligned}Y_i &= D_i Y_{1i} + (1 - D_i) Y_{0i} \\&= D_i \left(Y_{0i} + \delta_1 + \delta_2 X_i \right) + (1 - D_i) Y_{0i} \\&= Y_{0i} + \delta_1 D_i + \delta_2 D_i X_i \\&= \alpha + \delta_1 D_i + \delta_2 D_i X_i + \beta X_i + u_i\end{aligned}$$

Heterogeneity

This final equation

$$Y_i = \alpha + \delta_1 D_i + \delta_2 D_i X_i + \beta X_i + u_i$$

suggests that we want $p(X_i)$ and $D_i p(X_i)$, i.e.,

$$Y_i = \alpha + \delta_1 D_i + \delta_2 D_i p(X_i) + \beta p(X_i) + u_i \quad (1b)$$

which yields

1. a **group-specific treatment effect** $\delta_1 + \delta_2 p(X_i)$ for each X_i
2. an **average treatment effect** $\delta_1 + \delta_2 \bar{p}(X_i)$

More flexibility

We motivated propensity scores with a desire to reduce dimensionality and estimate/choose/assume fewer parameters.

Adding $\mathbf{p}(\mathbf{X}_i)$ and $\mathbf{D}_i\mathbf{p}(\mathbf{X}_i)$ as covariates in a linear regression doesn't quite exhaust our potential for flexible/nonparametric estimation.

Blocking

Option 2: Block (stratify) on propensity scores.

1. Divide the range of $\hat{p}(\mathbf{X}_i)$ into K blocks (e.g., 0.05-wide blocks).
2. Place each observation into a block via its $\hat{p}(\mathbf{X}_i)$.
3. Calculate $\hat{\tau}_k$ for each block via difference in means.
4. Average the $\hat{\tau}_k$ using their shares of the sample, i.e.,

$$\hat{\tau}_{\text{Block}} = \sum_{k=1}^K \hat{\tau}_k \frac{N_{1k} + N_{0k}}{N}$$

Note Blocking is similar to NN/kernel matching using $p(\mathbf{X}_i)$ as distance

Choosing blocks

Blocking on propensity scores requires defining blocks.

One common route involves some iteration.

1. **Choose blocks.**
2. Check the **balance of the covariates** within each block.
 - If covariates are **not balanced**, then split your blocks and repeat.
 - If covariates are **balanced**, then stop.

Note: Keep multiple-hypothesis testing in mind. With many covariates and many blocks, you are bound to find statistically significant relationships—even if you are actually balanced.

Overlap

Blocking emphasizes our overlap assumption, *i.e.*, $\mathbf{0} < \Pr(\mathbf{D}_i | \mathbf{X}_i) < \mathbf{1}$.

If a block contains zero treated/control units, we cannot calculate $\hat{\tau}_k$.

Caution Logit can hide violations—it forces $\mathbf{0} < \hat{p}(\mathbf{X}_i) < \mathbf{1}$.

Common practice: Empirically enforce overlap:

- Drop control units with $\hat{p}(\mathbf{X}_i)$ below the minimum propensity score in the treatment group.
- Drop treated units with $\hat{p}(\mathbf{X}_i)$ above the maximum propensity score in the control group.

Weighting

Option 3: Weight observations by the inverse propensity score.

Q: How does weighting by $\mathbf{1}/\hat{p}(\mathbf{X}_i)$ make sense?

A: Recall the simple difference in means, *i.e.*,

$$\hat{\tau}_{\text{Diff}} = \bar{\mathbf{Y}}_{\text{T}} - \bar{\mathbf{Y}}_{\text{C}} = \frac{\sum_i \mathbf{D}_i \mathbf{Y}_i}{\sum_i \mathbf{D}_i} - \frac{\sum_i (1 - \mathbf{D}_i) \mathbf{Y}_i}{\sum_i (1 - \mathbf{D}_i)}$$

which we've discussed is biased due to selection into treatment, *i.e.*,

$$E[\mathbf{Y}_{0i} | \mathbf{D}_i = 1] \neq E[\mathbf{Y}_{0i}]$$

Weighting, justified

Suppose we know $p(\mathbf{X}_i)$ and we weight each **treated** individual by $1/p(\mathbf{X}_i)$

$$\begin{aligned} & E \left[\frac{D_i Y_i}{p(\mathbf{X}_i)} \right] \\ &= E \left[\frac{D_i (D_i Y_{1i} + (1 - D_i) Y_{0i})}{p(\mathbf{X}_i)} \right] \\ &= E \left[\frac{D_i Y_{1i}}{p(\mathbf{X}_i)} \right] \\ &= E \left(E \left[\frac{D_i Y_{1i}}{p(\mathbf{X}_i)} \mid \mathbf{X}_i \right] \right) \\ &= E \left(\frac{E[D_i \mid \mathbf{X}_i] E[Y_{1i} \mid \mathbf{X}_i]}{p(\mathbf{X}_i)} \right) \\ &= E \left(\frac{p(\mathbf{X}_i) E[Y_{1i} \mid \mathbf{X}_i]}{p(\mathbf{X}_i)} \right) \end{aligned}$$

$$= E\left(\frac{p(\mathbf{X}_i) E[\mathbf{Y}_{1i} \mid \mathbf{X}_i]}{p(\mathbf{X}_i)}\right)$$

$$= E\left(E[\mathbf{Y}_{1i} \mid \mathbf{X}_i]\right)$$

$$= E[\mathbf{Y}_{1i}]$$

Similarly, weighting **control** individuals by $1/(1 - p(\mathbf{X}_i))$ yields

$$E\left[\frac{(1 - \mathbf{D}_i)\mathbf{Y}_i}{1 - p(\mathbf{X}_i)}\right] = E[\mathbf{Y}_{0i}]$$

Weighting: The estimator

Thus, we can estimate an unbiased treatment effect via

$$\hat{\tau}_{p\text{Weight}} = \frac{1}{N} \sum_{i=1}^N \left[\frac{D_i Y_i}{p(\mathbf{X}_i)} - \frac{(1 - D_i) Y_i}{1 - p(\mathbf{X}_i)} \right]$$

Intuition: We're trying to overcome selection bias, *i.e.*, treated individuals were more likely to be treated as a function of $\mathbf{X}_i$ — producing higher $p(\mathbf{X}_i)$ (consequently lower values of $p(\mathbf{X}_i)$ for control observations).

We want to get back to *as-good-as random* variation in treatment.

So we place more weight on:

(1) treated individuals with low $p(\mathbf{X}_i)$ and

Weighting: The example

Suppose for some individual i , $p(\mathbf{X}_i) = 0.80$.

This propensity score says someone with this set of $\mathbf{X}_i$ was four-times more likely to be **treated** than control.

Our weights fix this imbalance for each $\mathbf{X}_i$.

- If i is **treated**, then her weight is $1/p(\mathbf{X}_i) = 1/0.80 = 1.25$
- If i is **control**, then her weight is $1/(1 - p(\mathbf{X}_i)) = 1/(1 - 0.80) = 5$

And guess what $5/1.25$ is...

4!

This weighting scheme gets us back to equal representation for each set of $\mathbf{X}_i$.

Weighting: Last issue

Practical issue: Nothing guarantees $\sum_i \hat{p}(\mathbf{X}_i) = 1$.

Solution: Normalize weights by their total sum.

Applying the normalized (and estimated) propensity scores

$$\hat{\tau}_{p\text{Weight}} = \sum_{i=1}^N \frac{\frac{\mathbf{D}_i \mathbf{Y}_i}{\hat{p}(\mathbf{X}_i)}}{\sum_i \frac{\mathbf{D}_i}{\hat{p}(\mathbf{X}_i)}} - \sum_{i=1}^N \frac{\frac{(1 - \mathbf{D}_i) \mathbf{Y}_i}{1 - \hat{p}(\mathbf{X}_i)}}{\sum_i \frac{(1 - \mathbf{D}_i)}{1 - \hat{p}(\mathbf{X}_i)}}$$

Hirano, Imbens, and Ridder (2003) suggests this estimator is efficient.

Why choose one?

There's nothing special about weighted averages—regression can weight.

Thus, a **regression-based estimate**

$$Y_i = \alpha + X_i\beta + \tau D_i + u_i$$

with **weights**

$$w_i = \sqrt{\frac{D_i}{\hat{p}(X_i)} + \frac{(1 - D_i)}{1 - \hat{p}(X_i)}}$$

offers a *doubly robust* property—you have two chances to be right: $p(X_i)$ or the regression specification.

Why choose one? Part two

An alternative, doubly robust method combines propensity-score blocking with regression.

.note[Step 1] For each block k , we run the regression

$$\mathbf{Y}_i = \alpha_k + \mathbf{X}_i \beta_k + \tau_k \mathbf{D}_i + u_i$$

.note[Step 2] Aggregate block-level treatment-effect estimates

$$\hat{\tau} = \sum_{k=1}^K \hat{\tau}_k \frac{N_{1k} + N_{0k}}{N}$$

Major requirements

Don't get (too) caught up in the bells and whistles.

We still have two **major** requirements for any of these methods to work?

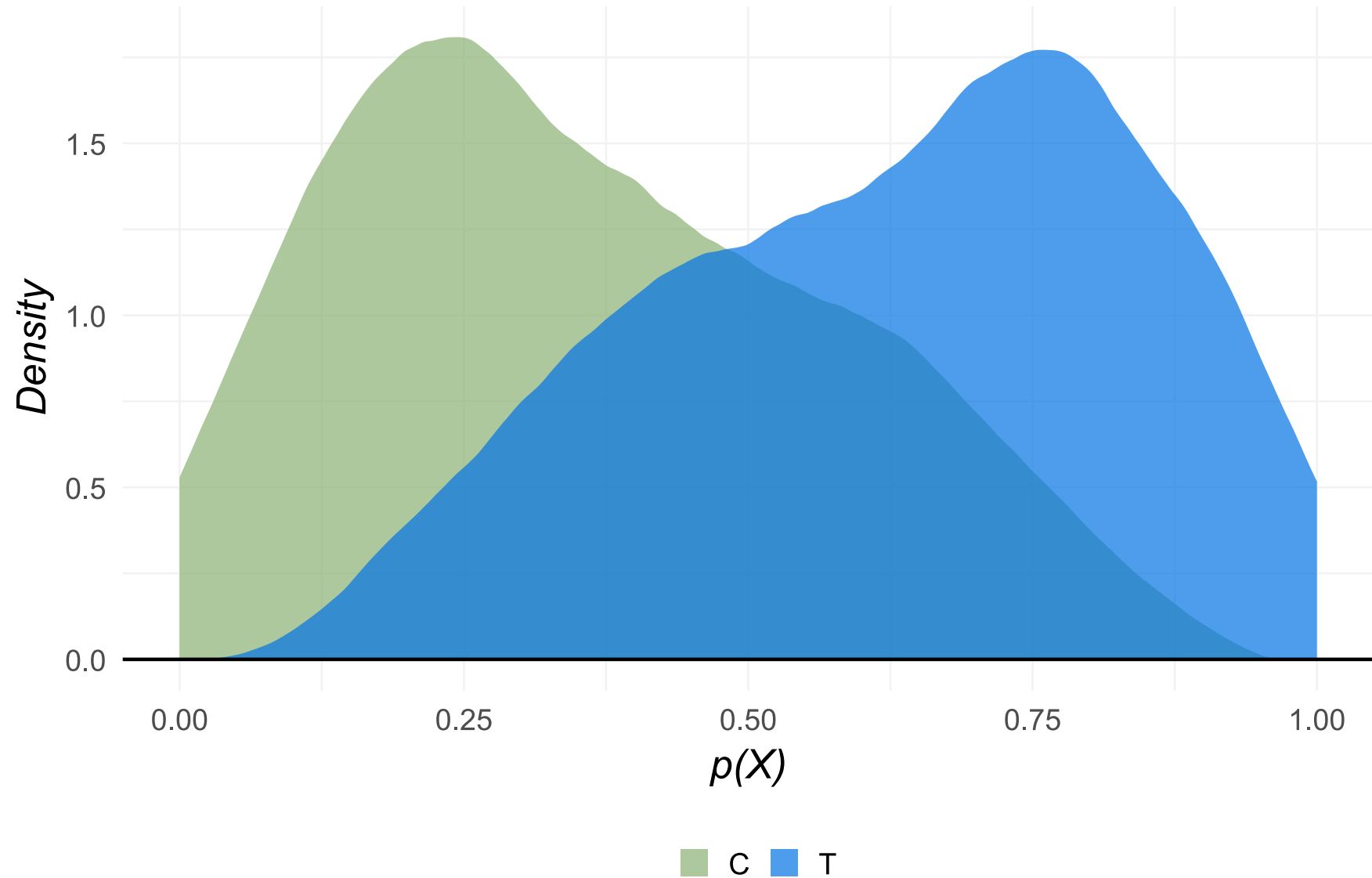
1. Is the **conditional-independence assumption** true?
2. Do we have **overlap** between treatment and control units?

We can look for evidence of (2) in the data—particularly if we're using propensity-score methods.

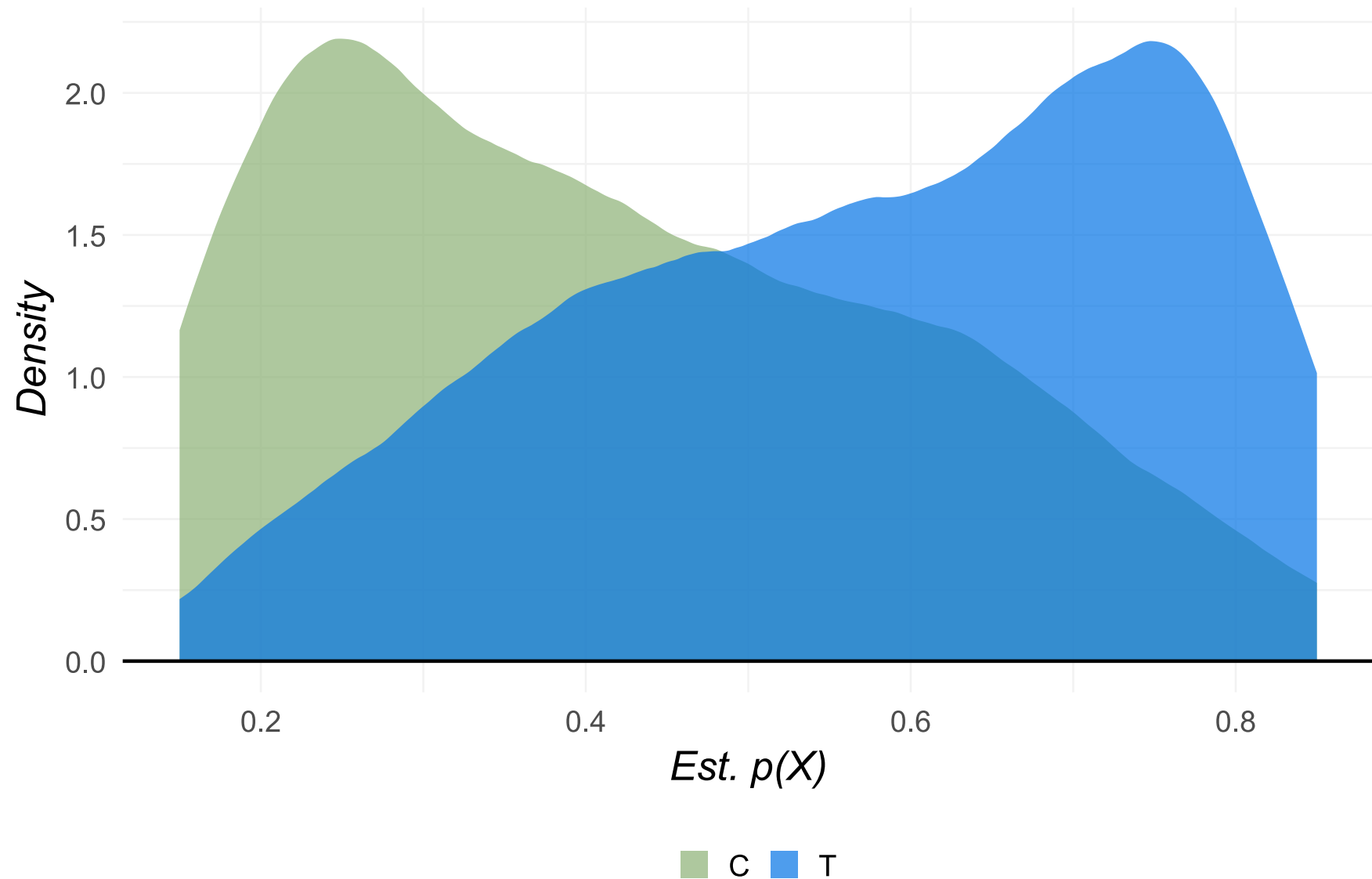
How? Plot the distributions of $p(\mathbf{X}_i)$ for **T** and **C**.

Note: Checking for overlap in **X**-space, can be tough as the dimensions of **X** expand. Careful of this when using logit to estimate the propensity score.

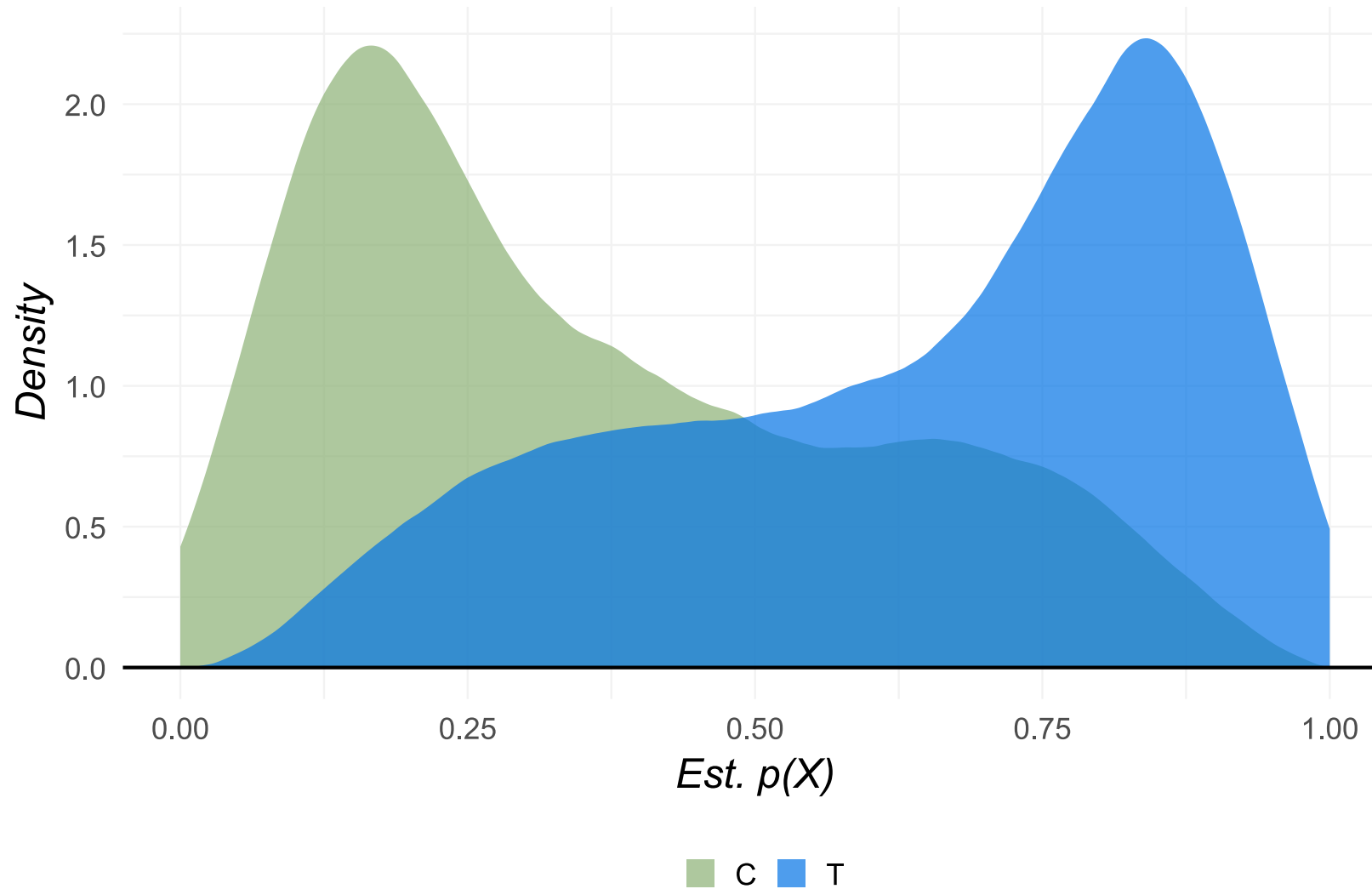
Missing overlap in $p(\mathbf{X}_i)$



Authentic (enforced) overlap in $p(\mathbf{X}_i)$

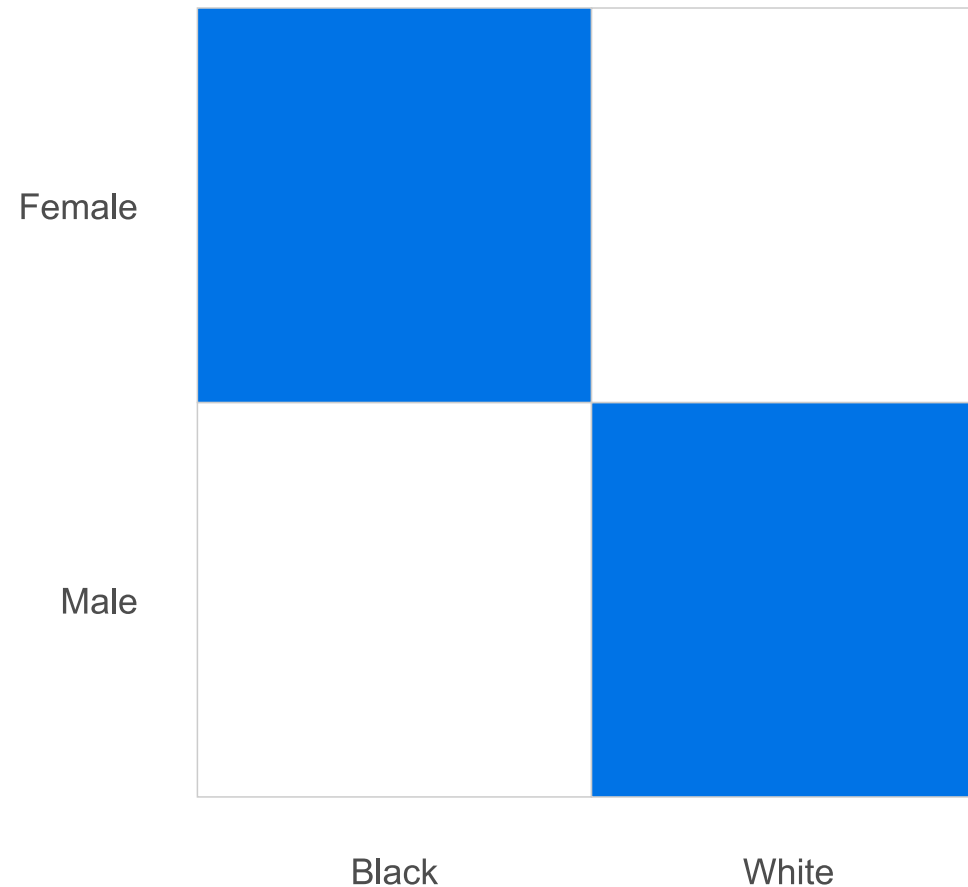


Logit-based $\hat{p}(\mathbf{X}_i)$ hiding some of the missing overlap in $p(\mathbf{X}_i)$



Overlap in one dimension does not guarantee in two dimensions.

Note: Shading denotes **share of treatment**: white=0% and blue=100%.



Extensions

Balance on observeables

It's important to keep in mind that our goal is to defend the conditional independence assumption.

This is an *assumption*. We cannot test it explicitly; rather we need to make a good argument that it holds.

One important component of this is *balance on observables*. Are the covariates balanced between treatment and control?

We **cannot** test balance on unobserveables (we don't observe them), but maybe we can make an argument that:

- (a) We have a lot of important observeables that impact treatment and outcomes.
- (b) If we have balance on observeables we also have balance on unobsereables.

Genetic Matching

Enter genetic matching: [Diamond & Sekhon. \(2013\)](#) - [Matching](#) in R

Recall that when we discussed blocking we can adjust the size of the blocks based on covariate balance.

This is somewhat ad hoc and also may mask nonlinearities/heterogeneity if we focus on balances in sample means (i.e. t-tests)

Genetic matching uses an algorithm to iteratively search for samples based on optimizing some balance metrics.

You can select the metrics, but ideally you can use non-parametric techniques that balance on the full distribution of the covariates (i.e. KS tests).

This formalizes a potentially ad hoc approach to the goal of achieving balance, and consequentially defending the CIA.

Pre-processing data

We showed how matching can generate estimators of treatment effects or matching-like approaches can be used in regression.

The goal in all of these is similar and the right technique depends on the assumptions, properties of the estimator, and the setting.

Another approach is to pre-process data using a matching estimator, and then run use whatever regression techniques you would use. The basic intuition is that you build a balanced sample by matching (potentially with weights), and then you've eliminated at least part of the selection bias in the full sample.

This approach was developed by some political scientist ([Ho et al. \(2007\)](#)) - **MatchIt** in **R**.

Pre-processing data

We'll get into this in more detail when we discuss quasi-experimental design, in particular difference-in-difference estimators.

Matching, pre-processing, and propensity score can be useful tools when combined with other causal inference research designs.

More recently this has been used in economics and shown to be able to recover treatment effects from randomized experiments (**Ferraro & Miranda (2017)**).

Examples

Example 1: Single discrete variable

```
1  ### create some covariates and error terms
2  set.seed(5302024)
3  dat <- data.table(
4    x1 = sample(1:10,1000,replace=T), #discrete covariate
5    x2 = rnorm(1000,mean=0,sd=2.5), # continuous covariate
6    eta = rnorm(1000), # unobserved factor in assigning treatment
7    e = rnorm(1000) # unobserved factor in outcome
8  )
9
10 ### assign treatment dependent on the discrete (x1), but not continuous variable (x2)
11 dat[, p := (x1+eta)/10]
12 dat[p>1, p := 1]
13 dat[p<0, p := 0]
14 dat[, d := rbinom(n = 1000, size = 1, prob = p)]
15
16 ### assign outcome
17 tau <- 5 ## treatment effect (constant)
18 dat[,y := tau*d + 1*x2 + e]
```

Example 1: Overlap

```
1 ### assess treatment overlap on x1  
2 table(dat$d,dat$x1)
```

```
#>  
#>  
#>      1    2    3    4    5    6    7    8    9   10  
#>  0  87  80  67  57  48  37  26  18    7    3  
#>  1  12  18  31  45  52  65  67  85  88 107
```

Example 1: Regression

```
1 ### regression
2 tau.hat.reg <- lm(y ~ d + x1, data=dat) ## linear x1; no control for x2
3 tau.hat.reg2 <- lm(y ~ d + as.factor(x1), data=dat) ## saturated in x1; no control for x2
4 tau.hat.reg3 <- lm(y ~ d + as.factor(x1) + x2, data=dat) ## saturated in x1; control for x2
5
6 stargazer(tau.hat.reg,tau.hat.reg2,tau.hat.reg3,
7           type='text',
8           keep = 'd',
9           omit.stat = c('aic','f','ser')
10          )
```

```
#>
#> =====
#>               Dependent variable:
#> -----
#>               y
#>            (1)   (2)   (3)
#> -----
#> d             5.011*** 5.016*** 5.055***
#>              (0.209) (0.210) (0.077)
#>
#> -----
#> Observations   1,000   1,000   1,000
#> R2             0.478   0.480   0.930
#> Adjusted R2    0.477   0.475   0.929
#> =====
#> Note:          *p<0.1; **p<0.05; ***p<0.01
```


Example 1: Exact matching by hand

```
1  ## exact matching by hand
2  dat[d==1,mean.1 := mean(y),by=x1] ## mean of treatment within x1
3  dat[d==0,mean.0 := mean(y),by=x1] ## mean of control within x1]
4
5  ## this just gets rid of NA by treatment
6  dat[,mean.1 := mean(mean.1,na.rm=T),by=x1]
7  dat[,mean.0 := mean(mean.0,na.rm=T),by=x1]
8
9  ## generate difference in means
10 dat[,mean.dif := mean.1-mean.0]
11
12 ## assign weights based on density of x1 among treated
13 weights <- as.data.table(table(dat[d==1,x1])) ## get the counts of x1 among treated
14 setnames(weights,c('x1','weight')) ## rename
15 weights[,weight := weight/length(dat[d==1,d])] ## normalize by number of treated obs
```

Example 1: Exact matching by hand

```
1  ## pull out mean differences within each value of x1
2  tau_x <- unique(dat[,list(x1,mean.dif)])
3
4  ## merge with weights
5  weights[,x1:=as.integer(x1)]
6  tau_x <- merge(tau_x,weights,by='x1')
7
8  ## generate weighted average
9  tau_x[,mean.dif.weight := mean.dif*weight]
10 sum(tau_x$mean.dif.weight)
```

```
#> [1] 5.029058
```

```
1  summary(tau.hat.reg2)$coef[2]
```

```
#> [1] 5.016034
```

Example 1: Matching with **Matching** package

```
1 ### exact match
2 tau.hat.match.exact <- Match(Y=dat$y, Tr=dat$d, X=dat$x1,
3                               exact=T,
4                               estimand = 'ATT',
5                               BiasAdjust = F)
6
7 sum(tau_x$mean.dif.weight)
```

```
#> [1] 5.029058
```

```
1 summary(tau.hat.match.exact)
```

```
#>
#> Estimate... 5.0291
#> AI SE..... 0.37073
#> T-stat..... 13.565
#> p.val..... < 2.22e-16
#>
#> Original number of observations..... 1000
#> Original number of treated obs..... 570
#> Matched number of observations..... 570
#> Matched number of observations (unweighted). 16236
#>
#> Number of obs dropped by 'exact' or 'caliper' 0
```

Example 2: Discrete and continuous variable

```
1 ### create some covariates and error terms
2 rm(list=ls())
3 set.seed(5302024)
4 dat <- data.table(
5   x1 = sample(1:10,1000,replace=T), #discrete covariate
6   x2 = rnorm(1000, mean=0, sd=2.5), # continuous covariate
7   eta = rnorm(1000,mean=0), # unobserved factor in assigning treatment
8   e = rnorm(1000) # unobserved factor in outcome
9 )
10
11 ### make treatment dependent on the discrete (x1) and the continuous variable (x2)
12 dat[, p := (x1+x2+eta)/10]
13 dat[p>1, p := 1]
14 dat[p<0, p := 0]
15 dat[, d := rbinom(n = 1000, size = 1, prob = p)]
16
17 ### assess treatment overlap on x1
18 table(dat$d,dat$x1)
```

```
#>
#>      1  2  3  4  5  6  7  8  9 10
#> 0  84  72  71  54  49  30  17  19  17  8
#> 1  15  26  27  48  51  72  76  84  78 102
```

```
1 ### assign outcome
2 tau <- 5 ## treatment effect (constant)
3 dat[,y := tau*d + 1*x2 + e]
```

Example 2: Regression

```
1 ### regression
2 tau.hat.reg <- lm(y~d + x1, data=dat)
3 tau.hat.reg2 <- lm(y~d +x1 + x2, data=dat)
4
5 stargazer(tau.hat.reg,tau.hat.reg2,
6           type='text',
7           # keep = 'd',
8           omit.stat = c('aic','f','ser')
9 )
```

```
#>
#> =====
#>                Dependent variable:
#>                -----
#>                  y
#>                (1)          (2)
#> -----
#> d                7.725***    4.973***
#>                  (0.181)    (0.084)
#>
#> x1               -0.202***    -0.006
#>                  (0.031)    (0.013)
#>
#> x2                      1.008***
#>                      (0.014)
#>
#> Constant         -0.377**     0.091
#>                  (0.166)    (0.068)
#>
#> -----
#> Observations      1,000        1,000
```

Example 2: Matching Maholanobis

```
1 ### Maholanobis distance
2 tau.hat.match.mah <- Match(Y=dat$y, Tr=dat$d, X=dat[,list(x1,x2)],
3                             Weight=2,
4                             estimand = 'ATT',
5                             BiasAdjust = F)
6 # MatchBalance(d ~ x1 + x2, match.out = tau.hat.match.mah, nboot=100, data=dat)
7 summary(tau.hat.match.mah)
```

```
#>
#> Estimate... 5.2592
#> AI SE..... 0.23277
#> T-stat..... 22.594
#> p.val..... < 2.22e-16
#>
#> Original number of observations..... 1000
#> Original number of treated obs..... 579
#> Matched number of observations..... 579
#> Matched number of observations (unweighted). 582
```

```
1 summary(tau.hat.reg2)$coef[2,]
```

```
#>      Estimate Std. Error    t value    Pr(>|t|)
#> 4.97331579 0.08401983 59.19216952 0.00000000
```

Example 2: Propensity score

```
1 ### propensity score with nearest neighbor matching
2 logit <- glm(d ~ x1+ x2, family = binomial, data = dat)
3 dat[,pscore := logit$fitted.values]
4
5 tau.hat.match.ps <- Match(Y=dat$y, Tr=dat$d, X=dat$pscore,
6                           Weight=2,
7                           estimand = 'ATT',
8                           BiasAdjust = F)
9 # MatchBalance(dat$d~dat$x1,match.out = tau.hat.match.ps,nboot=100)
10
11 summary(tau.hat.match.ps)
```

```
#>
#> Estimate... 5.0312
#> AI SE..... 0.68055
#> T-stat..... 7.3928
#> p.val..... 1.4366e-13
#>
#> Original number of observations..... 1000
#> Original number of treated obs..... 579
#> Matched number of observations..... 579
#> Matched number of observations (unweighted). 693
```

```
1 tau.hat.match.mah$est
```

```
#>      [,1]
#> [1,] 5.259167
```

```
1 summary(tau.hat.reg2)$coef[2,]
```

```
#>      Estimate Std. Error    t value    Pr(>|t|)
#> 4.97331579  0.08401983 59.19216952 0.00000000
```

Example 2: Propensity score (augmented)

```
1 ### propensity score, augmented
2 tau.hat.match.ps2 <- Match(Y=dat$y, Tr=dat$d, X=dat[,list(x1,x2,pscore)],
3                             Weight=2,
4                             estimand = 'ATT',
5                             BiasAdjust = F)
6 summary(tau.hat.match.ps2)
```

```
#>
#> Estimate... 5.3616
#> AI SE..... 0.2831
#> T-stat..... 18.939
#> p.val..... < 2.22e-16
#>
#> Original number of observations..... 1000
#> Original number of treated obs..... 579
#> Matched number of observations..... 579
#> Matched number of observations (unweighted). 580
```

```
1 summary(tau.hat.match.ps)
```

```
#>
#> Estimate... 5.0312
#> AI SE..... 0.68055
#> T-stat..... 7.3928
#> p.val..... 1.4366e-13
#>
#> Original number of observations..... 1000
#> Original number of treated obs..... 579
#> Matched number of observations..... 579
#> Matched number of observations (unweighted). 693
```


Example 2: Regression with propensity score

```
1 ### reg with pscore
2 tau.hat.reg.ps <- lm(y ~ d + pscore, data=dat)
3 tau.hat.match.ps$est
```

```
#>           [,1]
#> [1,] 5.031201
```

```
1 tau.hat.match.ps$se
```

```
#> [1] 0.6805541
```

```
1 stargazer(tau.hat.reg,tau.hat.reg2,tau.hat.reg.ps,
2           type='text',
3           omit.stat = c('aic','f','ser')
4           )
```

```
#>
#> =====
#>               Dependent variable:
#>               -----
#>               y
#>               (1)      (2)      (3)
#> -----
#> d               7.725***   4.973*** 5.023***
#>                (0.181)   (0.084) (0.190)
#>
#> x1             -0.202***   -0.006
#>                (0.031)   (0.013)
#>
#> x2                     1.008***
#>                      (0.014)
#>
#> pscore                     4.499***
#>                      (0.277)
```

```
#>  
#> Constant      -0.377**    0.091   -2.535***
```

Final Word

In general relying on the CIA - via control variables in a regression or with some form of matching - is a hard assumption to defend.

With high resolution fixed effects it is often *more* defensible.

However, a specific research design (identification strategy) really needs to tell the story of how you have random or quasi-random variation in your variable of interest.

Just saying, “I used matching so I have a causal effect” typically won’t fly in economics journals these days.

Think about this for your own research.